

Climate Data Analysis Bootcamp

Course Description

An introduction to modern Python tools for climate and geoscience data analysis using a selected dataset throughout the course. Students learn practical workflows for scientific computing, visualization, and multidimensional gridded data analysis.

Learning Outcomes

- Navigate JupyterHub/JupyterLab
- Use essential Linux commands
- Analyze data with NumPy, Pandas, and Matplotlib
- Work with NetCDF data using Xarray
- Apply resampling, grouping, rolling means, and interpolation

Day 1: Climate Data, Jupyter, and Linux Basics

- Climate data sources: observations, reanalysis, models
- NetCDF structure and dimensions
- LEAP Pangeo and Jupyter notebooks
- Introduction to Linux: Basic Linux commands, Navigating Files and Directories, working with Files and directories,
- Basics of Linux shell scripting
- Mini project: inspect and organize dataset

Day 2: Scientific Python Fundamentals

- Python basics: variables, loops, functions
- NumPy arrays
- Matplotlib plotting
- Pandas tables and time series
- Mini project: arrays and visualization

Day 3: Xarray for Gridded Climate Data

- Dataset/DataArray fundamentals
- Selection with isel and sel
- Aggregation and climatology
- groupby, resample, rolling, coarsen
- Interpolation and tips/tricks
- Mini project: monthly means and extremes

Software

- Python, JupyterLab, NumPy, Matplotlib, Pandas, Xarray, netCDF4 (CDO optional)

Recommended Platform

- LEAP Pangeo or any JupyterHub environment.
 - <https://leap.2i2c.cloud/>

References:

- <https://earth-env-data-science.github.io/intro.html>